MONTREAL.AI DIGITAL-20261027-R1

AGI ALPHA FLYWHEEL

When Capability Creates Opportunity

From a numerical failure to a verified digital answer

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

FIND YOUR ALPHA. BUILD YOUR SUCCESSOR Ω.

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27 October 2026 | Research review edition

Three live agent workflows. One public numerical mission.
64 fresh checks per workflow. Three complete passes.

Useful digital construction established within the frozen scope. General Operational Alpha, novel numerical capability and realized commercial return are not established.

A useful answer, with its limits attached

A public numerical failure can now be turned into an inspectable, executable answer within a bounded agent mission. This edition delivers a scalar beta-quantile compatibility kernel, a command-line interface, three frozen implementations, independent numerical checks, a complete replay package and an explicit commercial decision. It changes what a scientific software team can do immediately: reproduce an extreme-tail failure, obtain a checked replacement value within a stated domain, and inspect the evidence before deciding whether to integrate the component.

The concrete example matters. For probability 10^{-16} and beta shape parameters $a = 0.5$, $b = 3$, the installed SciPy 1.17.0 environment returned a quantile of **0.5**. The independently evaluated value is approximately $2.844444444444444 \times 10^{-33}$. The public report supplied the failure and a candidate numerical recipe. We did not discover that formulation. We converted it into bounded, failure-aware artifacts and tested the contribution of different ways of operating the available assistant. (chrisb83 2026)

All three workflows passed the **64 fresh confirmation checks**: 56 numerical or endpoint cases and eight required refusals. Ordinary competent operation and expert operation also handled a separately disclosed, already-known subnormal rounding trap; the original conventional implementation failed that diagnostic. Its correction is preserved separately. The fresh bank is one mission's case bank, not 64 independent missions or customers.

> **Decision: retain and deliver the useful engineering; pursue only a customer-funded qualification pilot. Do not promote this edition as an exclusive numerical breakthrough, a measured general Operational Alpha advantage, or a realized profitable business.**

Three verdicts

Question	What this edition establishes	What remains missing
Commercial delivery	An executable digital artifact accepted by the frozen research contract; a conditional service model with explicit thresholds	External buyer acceptance, willingness to pay, commitments, collections and complete actual costs
Operational Alpha	Live, tool-using implementations by ordinary and expert agents; both pass the fresh bank and the rounding diagnostic	A demonstrated improvement in accepted quality or complete economics from expert operation
Transformational contribution	A useful bridge from a reported failure to a bounded executable outcome	New numerical capability beyond the strongest identified established methods; an independent discovery comparison

The program's ambition remains larger than this result. Its discipline is to preserve a useful construction without borrowing an achievement from a failed hypothesis. The most productive successor may begin with a reliable piece of engineering and a sharper question, rather than with a claim of superiority.

The work was AI-assisted under the requested authorship. External authors retain credit for the report, recipe, libraries and foundational methods. No external peer review, production adoption or paid customer approval is implied.

The opportunity and the early challenge

Who benefits, and what changes

The immediate beneficiary is a maintainer or scientific software team evaluating a beta-tail calculation. Such a recipient needs more than a plausible formula: a reproducible failure, a bounded replacement, evidence for extreme and invalid inputs, and clear integration limits. Wrong finite values are particularly troublesome because a program may continue without an exception. Several observed failures emitted warnings; an application that ignores them can still consume an incorrect finite value. Whether a particular recipient's downstream decision is materially affected remains a customer-specific question; this study does not invent a deployed failure or financial loss.

The useful opportunity is **rapid numerical incident qualification**: assembling a precise failure report and a tested compatibility component from public inputs before an upstream release or a local migration is available. The changes making this operationally plausible are accessible source and issue histories, current numerical libraries, arbitrary-precision arithmetic and an assistant capable of reading code, compiling kernels, diagnosing failed checks and packaging evidence. This study observes that combination working locally. It does not establish that organizations were previously incapable of doing the same work.

Our shortlist was concentrated in scientific software, not a claim to an exhaustive cross-industry search. The preceding gap sprint had already explored broader opportunities. We narrowed the continuation to deliverable public incidents where the reasoning and execution were real and physical fabrication was unnecessary.

Candidate	Unmet outcome and beneficiary	Strongest early alternative and disposition
Beta-tail failure qualification	Checked scalar answers and an inspectable compatibility kernel for a numerical software team	Current Boost.Math plus ordinary high-precision methods solved the public examples. Selected for delivery and agent comparison; algorithmic novelty rejected early.
Boundary confidence intervals	Reliable boundary behavior and a validated repair for statistical software users	Existing active repair PRs already owned the relevant statsmodels issues. Excluded rather than duplicate ongoing work.
Kronecker-product correctness	A usable symbolic correction and regression evidence for SymPy users	An active repair PR already addressed the report. Excluded from construction.

The exact issue and PR records, rejected candidates and initial search notes are included in the evidence package. Statsmodels issues 10536/10538 had active PRs 10537/10539; SymPy issue 30860 had active PR 30863 at inspection. (statsmodels contributors 2026; SymPy contributors 2026)

Conventional falsification came first

Before the three agent constructions began, a separate workstream compiled current Boost.Math at commit `70a36632cfbfe6a862cd8a5b88b357ca3a770a3d`. A double-precision policy disabling promotion, with safe error handling and 400 root iterations, solved **all five public examples**. Its maximum relative error against the 80-digit public-recipe reference was approximately 2.008×10^{-14}. Raising the cap to 2,000 did not change that five-case result. A long-double configuration failed one example at both caps; the failed configuration remains in the record.

This is a meaningful established alternative. SciPy itself documents that `betaincinv` wraps Boost's `ibeta_inv`; the Boost interface already exposes precision and error policies. A successful current-

Boost configuration is therefore not an invented numerical method. (SciPy Developers 2026; Boost.Math Developers 2026)

Selection changed because of this evidence. The candidate survived as a useful digital delivery and an operational comparison. Its exclusive technical-breakthrough claim did not survive. Public-source formulation credit stays with the reporter; execution comparators received that formulation in full.

What was actually compared

Equivalent information, capable baselines

Three new agent workstreams received the same common prompt, public report and proposed recipe, pinned source extracts, conventional-challenge results, libraries, acceptance contract, tool access and resource limits. Each could inspect source, write code, compile a C++ kernel, run tests and recover from failures. Ordinary competent use was not reduced to a single prompt or forbidden from verification.

The expert arm received an explicit operating checklist: extract obligations, plan numerical regimes, verify independently, preserve failures and recover before freezing. The conventional arm was asked to compose the strongest practical established numerical techniques. Its implementation was built by an assistant too; it is **not a measured human-team trial**. The distinction is workflow and instruction, not a claim that conventional reasoning excludes tool-using AI.

All arms used the inherited available assistant configuration. The environment supplied the label **GPT-6 Astra Pro**. Provider-attested backend identity, temperature, sampling seed, token counts and billed cost were not exposed. These fields are recorded as unknown. No independent API endpoint was invoked to manufacture a stronger identity claim. The model work and local execution were live; this was not a simulation of successful model behavior. The result is narrower than a fully controlled, provider-metered same-model trial.

Computer execution consisted of shell commands, source inspection, compilation, numerical tests and file construction. No graphical interface performance was measured. No live `*.agent.agi.eth` identity, external settlement or production authority was asserted.

A bounded construction budget

The continuation reserved 120 minutes, starting at 18:06:05 on 27 October 2026 in Montréal, using existing authorized resources. Opportunity selection and conventional falsification preceded construction. Each implementation received a 15-minute, 25-tool construction ceiling with no subagents. The remaining continuation budget covered fresh confirmation, independent checking, economics, reproducible delivery and publication preparation.

Arm	Recorded construction time	Source freeze, UTC	Tool count as recorded
Ordinary competent	10.16 min	22:32:41	21
Expert operation	13.66 min	22:36:25	23
Conventional workflow	8.22 min	22:30:53	13

Tool-count conventions differ: the expert manifest includes progress messages, while the conventional count describes execution calls. They are traceability records, not a normalized efficiency measure. Timings start at the first recorded agent timestamp, not a provider scheduling boundary. The expert also recovered from a task-storage exhaustion during packaging; numerical work and original failed packaging evidence were preserved.

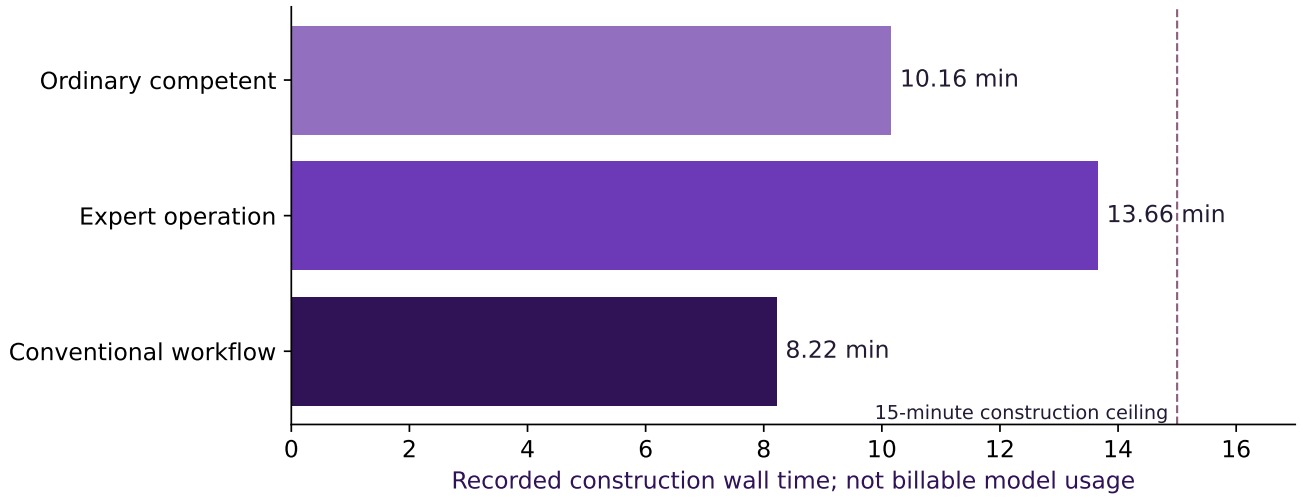


Figure 1: Recorded construction time. Expert operation took longer in this single trial. This is elapsed construction time, not token usage, billable cost or a population estimate.

Discovery and execution are separate experiments

No discovery advantage is claimed. The user supplied the program's objective; the continuation search found the public issue; the issue supplied the mathematical recipe. Every execution arm received it. A future discovery comparison would need equivalent initial information before revealing that recipe and a prespecified way to score the usefulness and originality of the formulation. A technical tie cannot supply that missing experiment retrospectively.

The accepted digital contract

The delivered object is deliberately bounded: **a scalar compatibility component**, not a drop-in replacement for all SciPy distributions, array broadcasting, accelerators, automatic differentiation or every positive beta shape.

A request contains an identifier and three numbers: probability q and shapes a, b. The interpretation is the exact binary64 value received by Python. The accepted numerical domain is

$$0 \le q \le 1, \qquad 0.25 \le a, b \le 10^6, \qquad \min(a, b) \le 100.$$

The result preserves the identifier and returns a status with a value or a refusal. Exact probabilities zero and one require their corresponding endpoint values. An interior true quantile that rounds to binary64 zero requires an explicit `underflow` status. Unsupported positive shapes are distinguished from invalid inputs. Booleans, strings, nonfinite values, nonpositive shapes and out-of-range probabilities are not silently coerced into numerical success.

For a verified reference value r strictly between zero and one, the frozen acceptance tolerance is

$$|\hat{x} - r| \le \max\{2\ \mathrm{ulp}(r),\ 10^{-11} \min(r, 1 - r)\}.$$

A representable positive reference may not be replaced with zero. Correct endpoint, underflow and refusal behavior is checked explicitly. A computation that cannot establish an answer returns `failed`; a plausible finite number does not automatically count as success.

How the construction works

The three implementations combine established instruments: a pinned Boost inverse, analytic special cases, incomplete-beta continued fractions, symmetry, log-domain arithmetic and high-precision safeguarded inversion when needed. This composition protects against numerical scale and unsupported inputs. Its scientific ingredients are conventional. Definitions, symmetry, the integer-shape binomial identity and continued fractions appear in NIST's reference treatment. (National Institute of Standards and Technology 2026)

The ordinary implementation is the recommended research entry point because it passed the primary contract and the disclosed rounding regression with less recorded construction time than the expert arm. That is a local delivery choice, not a universal ranking. All original implementations remain available so a recipient can inspect the tradeoffs rather than inherit a hidden selection.

For the public example, a recipient can run:

```
python experiment/arms/ordinary/run.py <<'JSON'
{"id":"beta-tail","q":1e-16,"a":0.5,"b":3.0}
JSON
```

The expected accepted value is approximately `2.844444444444444e-33`, with status `ok`. The package supplies a portable rebuild/replay command so a recipient need not trust the included platform-specific binary. Python and a C++17 compiler are required for that rebuild. Dependencies and licenses are pinned and included.

Acceptance does not transfer itself

The frozen numerical envelope is the product boundary. A recipient integrating the component into a different distribution, vector API, operating system, threat model or safety-critical process needs new qualification appropriate to that use. This is a concrete limitation, not an excuse to omit the delivered bounded result. External customer acceptance was not observed; research acceptance and customer acceptance remain separate records.

Fresh confirmation and independent checking

All three arm manifests were verified before the case bank was generated. The root source freeze was recorded at **22:40:20.056812 UTC**; generation followed at **22:40:20.103189 UTC**. The generator's recorded seed is `2707272985118872086`. The seed was drawn after source freeze, and the case bank was generated once. No primary implementation was changed after viewing these cases.

The bank contained 24 extreme-tail cases, 12 central cases, eight analytic-shape cases, four integer-shape cases, eight endpoint or underflow cases, and eight invalid or unsupported requests. It deliberately emphasizes adverse numerical regimes. It is not a random sample of customer demand or the prevalence of bugs in software.

The independent root evaluator used 80- and 120-digit arithmetic, a log-domain bisection inverse and a separately written incomplete-beta continued fraction. It did not call Boost's inverse as its primary judge. Analytic cases and exact rational integer-shape identities provided additional controls. All 64 references were resolved. A separate audit recomputed acceptance and used Decimal/Fraction controls; its complete check record is supplied.

What the checks do and do not prove

The bank establishes finite acceptance under a frozen contract. It does not prove correctness throughout the continuous parameter envelope. Two high-precision runs can share an error, and distinct programs can share a mathematical ingredient. The evaluator therefore retains its independent controls and uncertainty rather than treating agreement as a proof of universal correctness.

Every primary call met its observed CPU limits: five seconds per call and 120 seconds per worker.

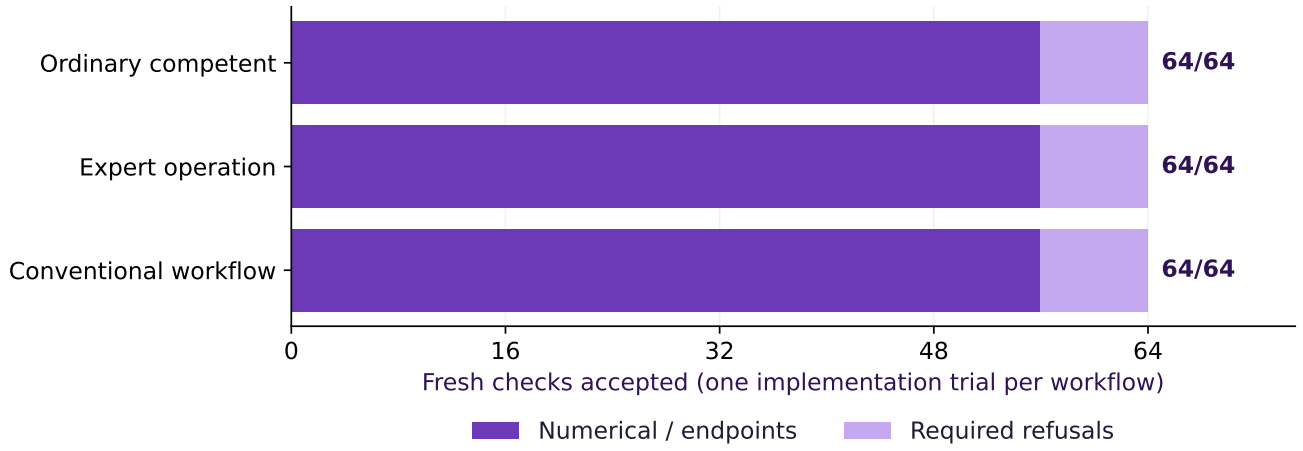


Figure 2: Primary fresh confirmation: all three frozen workflows satisfy 56 numerical or endpoint cases and eight refusals. The separately disclosed known midpoint diagnostic is outside this denominator.

Raw timings are retained, but **no numerical speedup is claimed**. Many recorded per-call CPU differences were zero; the tiny batch and observed timer behavior do not support a stable ratio. The identical reported peak memory high-water marks are likewise unsuitable for an arm-level memory advantage. Import and build time are recorded separately from solve time, and construction and verification effort cannot be replaced by a microsecond solve-time figure.

The broad result is uncomplicated: competent ordinary operation succeeded, explicit expert operation succeeded, and the strongest identified conventional workflow succeeded on the fresh bank. The expert checklist did not produce a better primary accepted outcome in this trial.

The additional boundary failure stays visible

After primary confirmation, the independent audit ran an already-known diagnostic at the smallest positive binary64 probability, with $a = 1, b = 2$. Ordinary and expert implementations returned the correct positive subnormal value; the original conventional implementation returned zero with `underflow`. This diagnostic was not fresh and was not added to the primary denominator. A separately labeled conventional correction, using ordinary analytic precision and lattice rounding, passed the same 64-case replay plus the known midpoint diagnostic. These are post hoc replays, not a new blind confirmation; the original failure remains in the correction record. Neither the original failure nor the corrected comparison disappears.

A better measuring instrument

The most instructive failure occurred in the numerical judge. Both the ordinary and expert workstreams independently noticed that converting a high-precision value to a machine float can conceal a subnormal rounding error. Agreement at two precisions did not guarantee correct rounding. The root evaluator initially shared that weakness.

The correction was made **before generating the fresh bank**. The old evaluator, new evaluator, source hashes, diagnostic and rationale remain in the package. The acceptance requirements did not change. All three primary arms were evaluated by the same corrected instrument. We disclose the feedback from the arm builders because the root evaluator was not sealed from all development feedback.

An exact counterexample

Let $\delta = 2^{-1074}$, the smallest positive binary64 value. For a Beta$(1, 2)$ distribution,

$$F(x) = 2x - x^2.$$

At the midpoint between zero and δ,

$$F(\delta/2) = \delta - \delta^2/4 < \delta.$$

The quantile at probability δ is therefore **above** that midpoint and rounds to δ, not zero. A computation that loses the tiny positive correction can land on the midpoint and apply the wrong rounding decision. This is a numerical instrument failure with an exact rational adjudication, not an invented new special function.

The corrected reference explicitly rounds on the binary64 subnormal lattice. Near a midpoint, it uses an exact integer-shape CDF when available, otherwise a 500-digit adaptive calculation with an unresolved-reference outcome if the decision remains too close. The construction uses conventional mathematical identities; the contribution is the preserved diagnosis and the accountable correction.

Why this belongs in the main narrative

A fluent implementation and a passing high-precision comparison can still be wrong in the same way. A useful agent must be able to question its checker and produce independently inspectable evidence for that challenge. In this study, **ordinary competent use already did that**. It would be misleading to award the behavior exclusively to the expert arm.

Transformational Intelligence argues that evaluators may themselves impose restrictions, but that revising an evaluator must not become a way for a candidate to award itself victory. Its prescribed discipline is an independent outcome anchor, preserved old evidence and a common corrected comparison. This continuation exercises that discipline on a real numerical instrument. It does not thereby establish a novel theory of evaluator revision. (Boucher 2026c)

The earlier kinetics construction remains supporting engineering. Its conventional parity, thin conditional economics and untested physical assumptions are unchanged. No chemical parameters, catalysts or acceptance criteria were tuned in this continuation to manufacture a second success.

The commercial offering and the cost boundary

The free delivered artifact is the recipient's strongest outside option. A prospective paid engagement must therefore purchase **bounded qualification, integration and handoff in one declared buyer environment**, including correction and support. It cannot earn a license premium for a public conventional algorithm. That buyer-specific engagement has not been performed or accepted by a customer; the following economics are conditional on it.

All amounts are nominal US dollars. Let M be the unknown additional model/tool cost per future engagement, including retries and failed attempts. Let D be the unknown complete historical cost of this research, development and publication project. Neither is inferred from wall time. The same M is used for all workflows because no metered cost advantage was established. Historical D is deducted once for full-project recovery, not again per customer.

Complete central cost per accepted engagement

Item	Explicit central assumption	Cost or receipt
Net revenue	$1,500 price, less 2% expected refunds	$1,470.00
Successful delivery	2.5 h at $125/h: scope, integration, qualification, handoff	$312.50
Failed attempts	20% failure probability; 1.5 h each; expected 0.375 h per acceptance	$46.875
Support and correction	0.5 h at $125/h	$62.50
Acquisition	0.75 h at $125/h plus $25 cash	$118.75
Infrastructure	Per accepted engagement	$5.00
Payment expense	Assumed 3% of price plus $0.30	$45.30
Additional model/tools	All future engagement retries included	M
Contribution	Before cohort fixed costs	$879.075 - M$

No recurring human review is invented for each automated scalar calculation. These hours describe future buyer integration and service, not observed labor in the present experiment. Loaded rates of $100–175/h are scoped estimates; historical employee wages provide only an order-of-magnitude anchor, not contractor quotations. The complete cost model discloses hours, rates and uncertainty. (US Bureau of Labor Statistics 2026)

Four central buyers share $800 of maintenance and fixed business cost over the cohort; $500 of additional commercial setup is charged once. That setup concerns service contracting and onboarding, not rebuilding the already delivered code, tests or documentation. The central cohort extends through day 115. The favorable cohort extends through day 195; neither is an annual forecast.

Scenario	Buyers and price	Operating profit	Full-project return
Favorable	8 at $2,500	$16,526.93 - 8M$	$16,226.93 - 8M - D$
Central	4 at $1,500	$2,716.30 - 4M$	$2,216.30 - 4M - D$
Adverse	No buyers	$-1,650$	$-3,050 - D$

At an illustrative $M = \$100$, central operating profit is $2,316.30; full-project return is $1,816.30 - D$. This is a sensitivity coordinate, not a measured model invoice. Operating profit requires $M < \$679.075$. At $M = \$100$, the operating break-even price is $890.45; recovering additional setup raises it to $1,022.03 before historical D. Full-project profitability remains unknown.

The adverse case preserves zero demand rather than forcing a sale. A conditional-commitment gate can avoid its prospective setup and fixed-cost loss. If an adverse low-price sale did occur, its unit contribution would be negative even before model cost. Expected profit is not an observed earning.

Financing, demand and the next voluntary exchange

Buyer value must survive the free alternative

The central price hypothesis has little room for error. It assumes eight hours of avoided self-qualification at $150/h, $600 of avoided delay, and one hour of buyer participation. The resulting value ceiling is **$1,650**, leaving only **$150 of modeled buyer surplus** at the proposed $1,500 fee. The hours and delay value are unmeasured. A competent recipient needing only two hours to reuse the free artifact, with no valuable delay avoided, may rationally buy nothing.

Prices are independent buyer hypotheses, not a fixed discount from an invented market price. The executable sensitivity grid holds volume fixed only to show arithmetic; it does not assume buyers remain when price exceeds their value. Extra participation is not automatically beneficial: replacing a $125 provider hour with a $150 buyer hour increases combined cost unless it brings another justified benefit.

A human team attempting an equivalent new repair from the same public recipe is estimated separately at 14–38 hours, or $1,400–$6,650 at the stated rate endpoints. The central attempt estimate is 24 h at $125/h. This is neither a measured human baseline nor willingness to pay, and it is never added as a second bill for the already produced artifact.

Cash comes before collection

At central $M = \$100$, acquisition and pre-acceptance work precede collection. The modeled peak expected cash deficit is **$2,091.25**. A separate **$1,875 reserve** covers one additional full refund and three extra correction hours. Required unrestricted funding is therefore **$3,966.25**. Actual available funds, reserve balances, credit and financing commitments are unknown. Any unpaid historical development obligation must be added.

An earned milestone can reduce the funding threshold to **$3,476.55**: 30% is paid after a separately useful, accepted qualification package; the remainder follows final accepted delivery. This arrangement preserves the numerical requirements, refund exposure and reserve. Two collection fees reduce central full-project return by $1.20 to $1,815.10 - D$ at the illustrative model-cost coordinate; the initial calculation omitted that second fee and is preserved in the economic correction record. It is a conditional contract design, not an observed customer agreement or legal approval. A refundable deposit stays restricted and supplies **zero unrestricted operating capital**. A conditional commitment alone supplies no cash.

The reserve covers the specified stress, not multiple correlated failures. Two extra full refunds exceed its refund component by $1,500. Delayed collection and financing fees can consume the margin. A reserve is cash held, not an operating expense; it is not charged twice.

Sharing is available to the conventional team too

Matched arrangement, central $M = \$100$	Return before historical D
Conventional, four isolated engagements	$-2,083.70$
Expert operation alone, isolated	$-2,083.70$
Conventional, shared fixed costs	$1,816.30$
Expert operation plus shared fixed costs	$1,816.30$

This accounting counterfactual assumes equal capability and equal costs, as the measured comparison warrants. The interaction attributable to expert operation is zero. Sharing a known fixed cost can make a conditional service viable; it is not a novel institution or exclusive Alpha. All buyers retain the same public rights and can decline the service.

Observed commitments and collections: zero. Complete actual project cost and available funds: unknown. Buyer value, scope and costs still need verification.

What the result makes possible

Evidence: a small capability that works now

The demonstrated near-term capability is a bounded numerical incident package: locate the relevant public evidence, reproduce the failure, challenge the proposed novelty, construct a usable scalar component, uncover a checker weakness, run fresh confirmation and deliver the complete record. Its usefulness is inspectable without waiting for physical fabrication, private data or a speculative future model.

The external libraries and public report do much of the intellectual work. The live assistant's contribution is to connect them into an executable, checked outcome under a time limit. That is a real productive capability, even when a competent conventional team could perform the same task. The study does not establish that the expert operating procedure improves that capability over ordinary competent use.

Conditional expansion: a portfolio of qualification missions

A broader service could qualify numerical incidents across scientific packages, supply bounded compatibility components and help recipients decide whether to integrate, upgrade or decline a workaround. That expansion depends on evidence absent from this single mission: multiple prospectively selected incidents, real recipient acceptance, adequate model metering, known licensing and support obligations, and a measured rate of failures that cannot be repaired within the paid scope.

The next economically informative step is one voluntary buyer pilot with a written acceptance contract and a capped integration scope. A valuable customer question might be whether this component resolves a blocked calculation on the buyer's actual supported stack. It must not be silently replaced by passing our existing numerical bank. The free artifact remains the buyer's outside option; a buyer who can self-integrate cheaply has no reason to pay a premium merely because an agent was involved.

Long-term ambition: capabilities that open better opportunities

The flywheel's larger thesis is that a delivered capability can make a later opportunity feasible. This result supplies a possible starting asset, not evidence of compounding returns. A future successor could reuse the source-pinning, numerical-regime analysis or exact-midpoint checks on a genuinely new mission. It would still have to beat its own competent baseline and an ordinary library containing the same asset after retention, validation, integration and maintenance costs.

No fresh transfer trial was performed here. There is no retained-Alpha or Gym-training claim. Adding transfer work would not repair the missing expert advantage, and ordinary reuse is already available. Optional learning investment should follow an expected benefit, not a prescribed number of reinvestments.

Invent the Economy separates technical feasibility, participant value, profitability, liquidity and actual exchange. Transformational Autonomous Enterprises preserves comparisons in which a supplied deterministic specialist matches quality and costs less. Those lessons shape this edition: keep useful assets, let strong baselines win, and require evidence for the next economic claim. (Boucher 2026a, 2026b)

Recommendation: deliver the research artifact, retain the corrected numerical instrument and conventional comparisons, and invite a narrowly scoped customer qualification pilot. Stop the exclusive algorithmic-novelty claim. Leave general Operational Alpha, discovery advantage, retained benefit and realized commercial viability unestablished until their own experiments support them.

Reproducibility, provenance and publication

One edition, several readable forms

The package contains the compilable LaTeX manuscript, bibliography, vector figures and plotting source; the polished PDF; a concise executive brief; a complete offline HTML edition and a GitHub-readable manuscript; frozen prompts, protocol, source and fresh results; the conventional challenge and failures; the evaluator correction; independent audit; executable economics; and a LinkedIn draft that has not been posted.

The Markdown manuscript is the common content source for the LaTeX and HTML editions. Numerical facts are checked against the machine-readable result and economic files. The brief repeats the same verdicts and conditions. Download identities and the publication status are recorded separately so a review upload cannot be confused with a canonical release.

The deterministic reproduction entry point rebuilds kernels in isolated copies, replays the frozen case bank, recomputes references and scores, and preserves the original source hashes. The native bridge reuses the existing Journal, CostLedger and fitness components; it does not build a second registry or invent a customer payment. Unknown actual charges remain unknown. Its local research identity is not a live external authentication claim.

Evidence and source identities

Item	Identity or location
Edition	`DIGITAL-20261027-R1`
Repository audit source	`3947e4ec8fd74ad40e13e8ff825a149fa98a10e9`
Boost.Math construction source	`70a36632cfbfe6a862cd8a5b88b357ca3a770a3d`
SciPy inspected source	`37fad094584e1129ea652680608cd2c1b02463b9`
Acceptance protocol	`experiment/protocol.json` and SHA-256 sidecar
Fresh source and case freeze	`experiment/source_freeze.json`, `fresh_case_metadata.json`
Primary result	`experiment/comparison_results.json`
Instrument correction	`experiment/evaluator_correction.json`
Complete claim map	`delivery/claim_map.json`
Monetary model	`economics/economics.py` and scenario outputs

All 24 preserved upstream component files were verified identical at the prior and current repository snapshots. Third-party licenses remain attached to their sources. Frozen experimental originals are retained byte for byte; any publication-normalized or corrected copies have separate identities. The previous gap archive is preserved with its original SHA-256 in the continuation provenance.

Publication is a separate acceptance boundary

The authorized route is an additive study review branch and draft pull request under `studies/agi-alpha-digital-outcome-20261027-2206/`. The mixed source, data and HTML package takes the repository's full review lane. It is not a passive-paper exemption. Existing runtime, discovery, accounting and publication owners retain their roles. (MONTREAL.AI 2026a, 2026b)

This edition may be inspected and downloaded from its pinned review source. Canonical production publication requires the existing owner's exact-source review, required hosted checks, qualified integration and actual public deployment/served-byte verification. A green preparation workflow with skipped deployment does not establish publication. The delivery receipt states the exact source, public reading page, downloads and any remaining external dependency; it does not borrow a prior release's approval.

Appendix: preserved failures and claim boundaries

Corrections without erasure

Record	Observed problem	Treatment
Early Boost challenge	One long-double configuration failed a public example	Preserved beside the successful double configuration; no weak-only baseline
Ordinary development	Subnormal rounding and an unscaled quadrature oracle produced misleading evidence	Original attempts retained; lattice rounding and normalized controls added before freeze
Expert development	Wrong finite subnormal-tail outputs and high-precision conversion error	Trigger and rounding repairs made before freeze; original attempts retained
Root evaluator v1	Direct high-precision-to-float conversion could double-round	v1 retained; exact midpoint controls and v2 freeze before fresh generation
Conventional primary source	Known post-confirmation $\mathrm{Beta}(1,2)$ midpoint returned underflow	Failure retained outside primary denominator; correction separately labeled
Expert packaging	Storage exhaustion interrupted README creation	Partial attempt and recovery recorded; no numerical source erasure
Root freeze check	A hash copied into a summary contained a transcription error	Original source manifest rechecked; no source mutation or case generation occurred before resolution

What each positive result can support

- **Runnable construction:** yes, within the stated scalar envelope and finite tested cases. It is not universal proof or upstream integration.
- **Fresh research acceptance:** 64/64 checks for each original frozen arm, under a common corrected evaluator. Known diagnostic limitations remain visible.
- **Commercial operating profit:** conditional model only, with price, volume, scope, failure, support and unknown model-cost thresholds. No observed sale.
- **Financing feasibility:** conditional on actual unrestricted funds or explicitly agreed stages. Refundable balances do not become operating capital.
- **Operational Alpha:** not established. The expert arm did not improve primary accepted quality over competent ordinary use; complete cost and broad replication are missing.
- **Transformational contribution:** no exclusive numerical advance. Established methods meet the useful outcome; the public reporter supplied the formulation.
- **Fresh-transfer benefit:** untested. Ordinary reusable-library access remains a required comparator for a future transfer study.

Research limits that affect the headline

One implementation trial per arm on one public mission cannot establish population performance. Provider identity, billing and buyer demand are unmeasured. Independent implementation and audit occurred within one coordinated assistant session. Finite scalar acceptance does not justify broader product claims. The conventional midpoint defect and post hoc correction stay outside fresh confirmation. Human-service estimates are assumptions, not quotations or willingness to pay.

References

Boost.Math Developers. 2026. “The Incomplete Beta Function Inverses.” https://www.boost.org/doc/libs/latest/libs/math/doc/html/math_toolkit/sf_beta/ibeta_inv_function.html.

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